Maksat Rakhat

Laboratory work 4

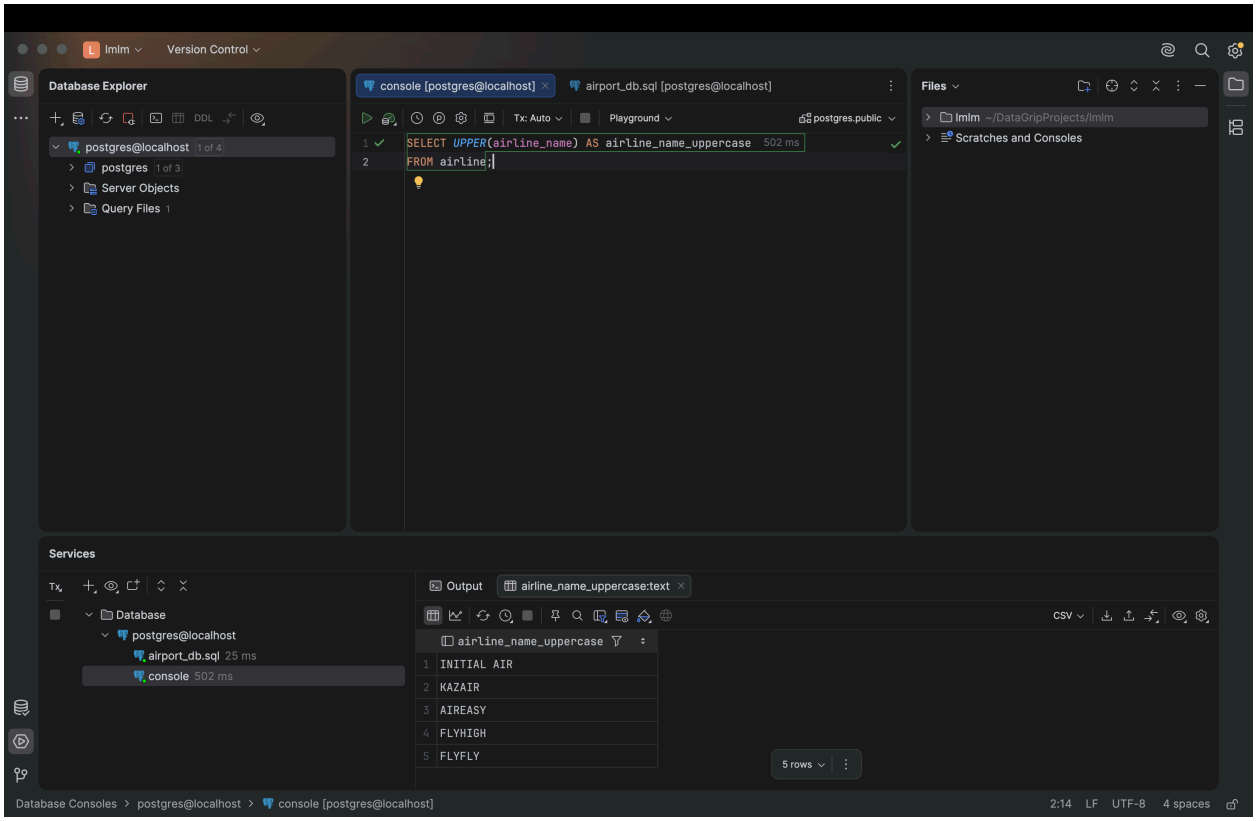
We continue to work with the database from the previous laboratory works.

Take a full-page screenshot that covers the code and results of each task.

Tasks:

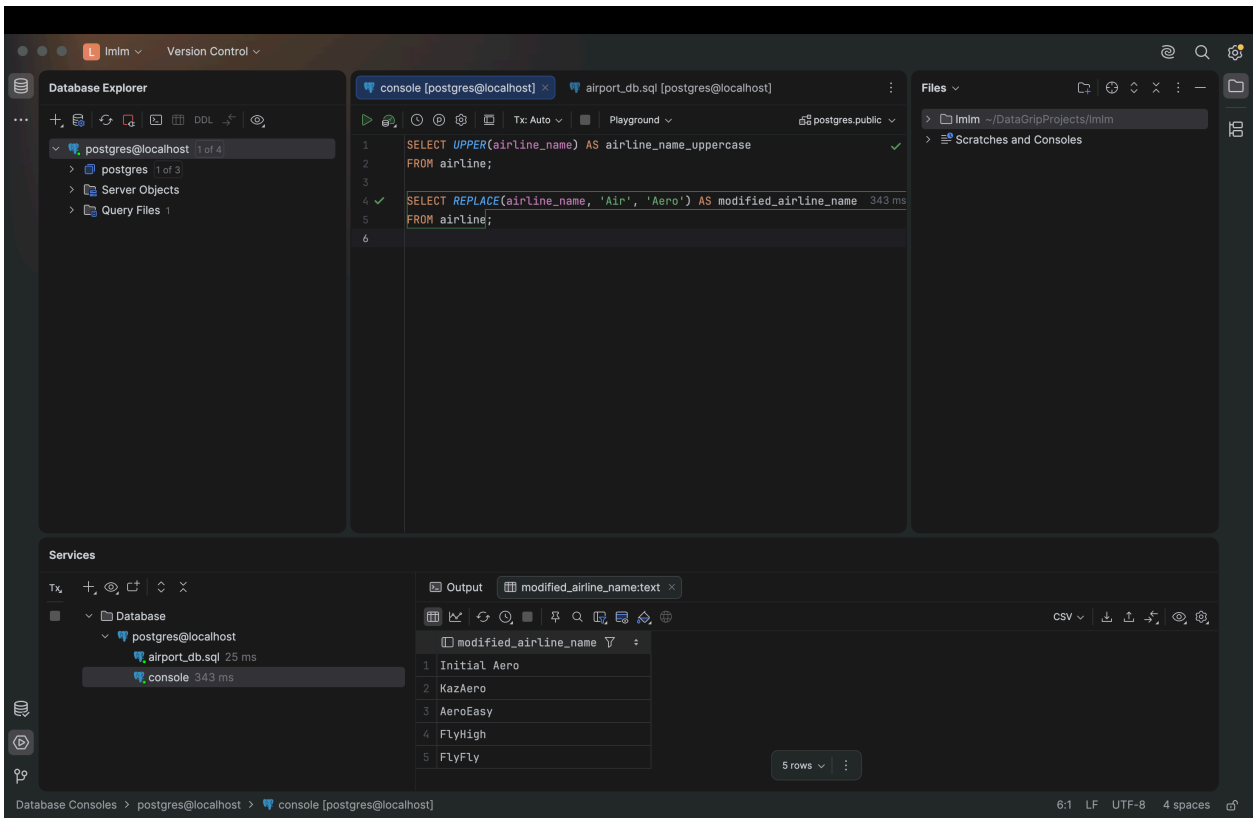
1. Retrieve all airline names in uppercase.

```
SELECT UPPER(airline_name) AS airline_name_uppercase
FROM airline;
```



2. Replace any occurrence of the word "Air" in airline names with "Aero".

```
SELECT REPLACE(airline_name, 'Air', 'Aero') AS modified_airline_name
FROM airline;
```



3. Fing all flight numbers that coordinates with both airline 1 and airline 2.

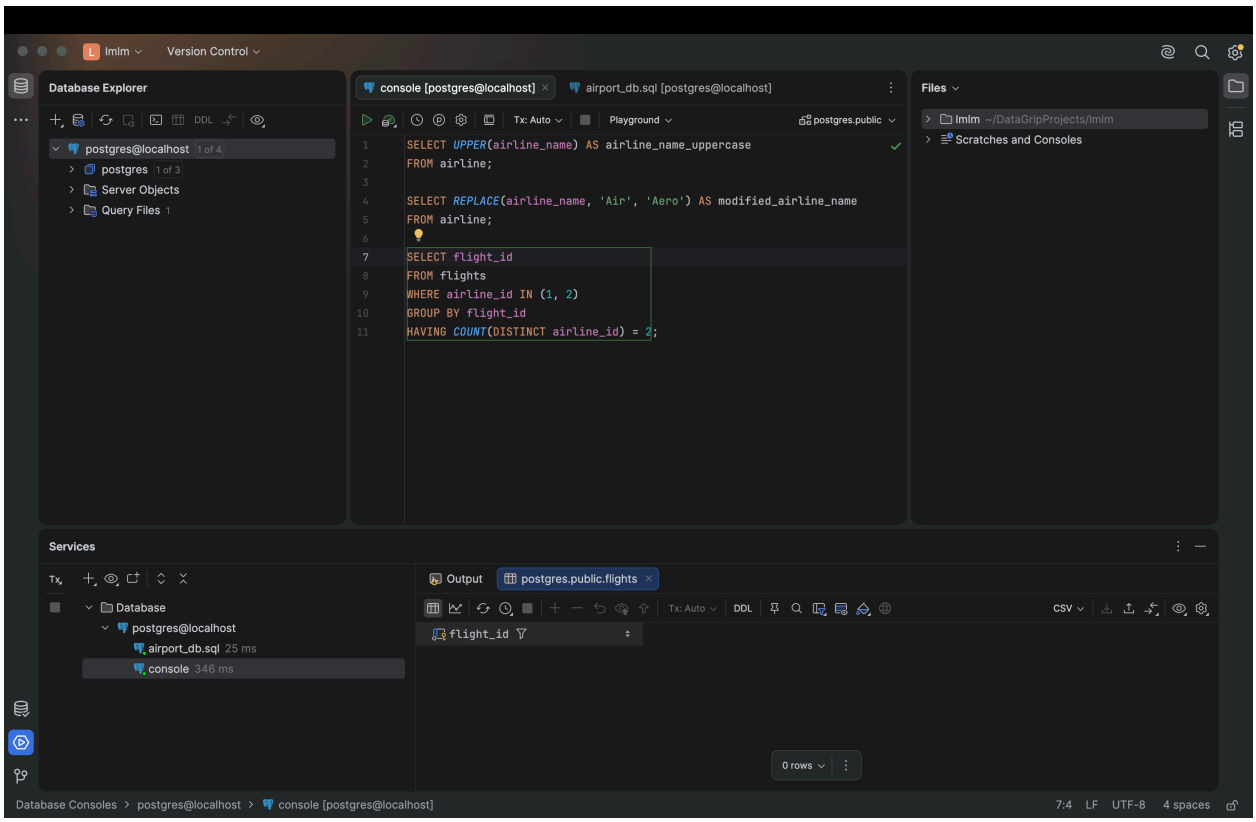
```
SELECT flight_id

FROM flights

WHERE airline_id IN (1, 2)

GROUP BY flight_id

HAVING COUNT(DISTINCT airline_id) = 2;
```



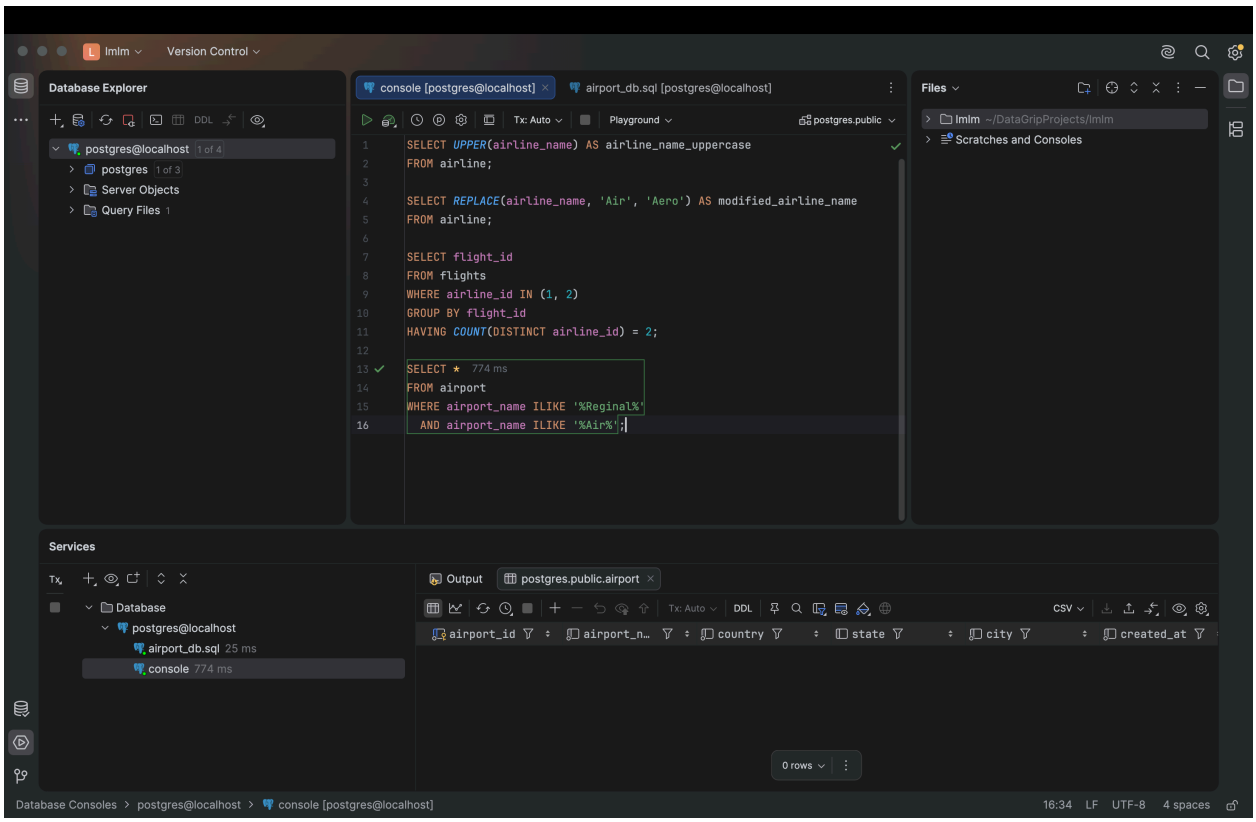
4. Retrieve airports that contain the word "Reginal" and "Air" in their names.

```
SELECT *

FROM airport

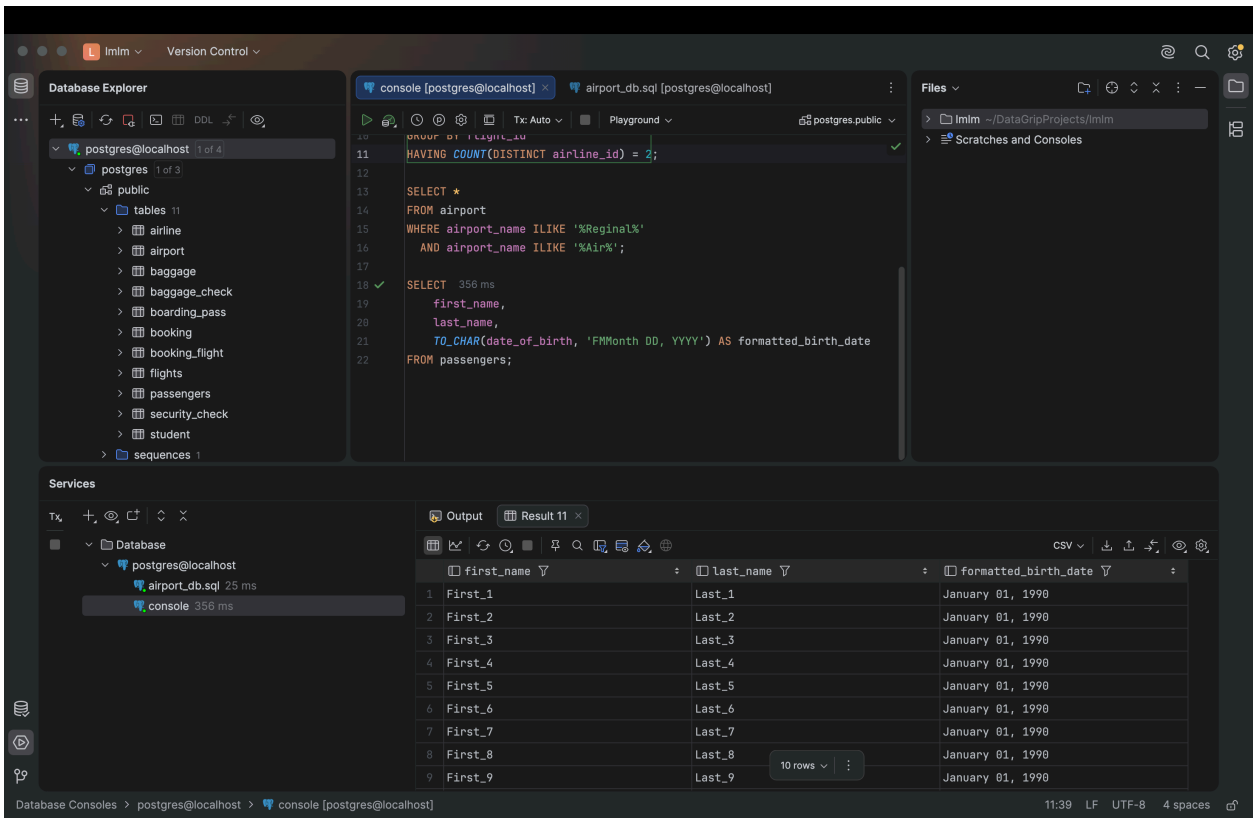
WHERE airport_name ILIKE '%Reginal%'

  AND airport_name ILIKE '%Air%';
```



5. Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

```
SELECT
    first_name,
    last_name,
    TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS formatted_birth_date
FROM passengers;
```

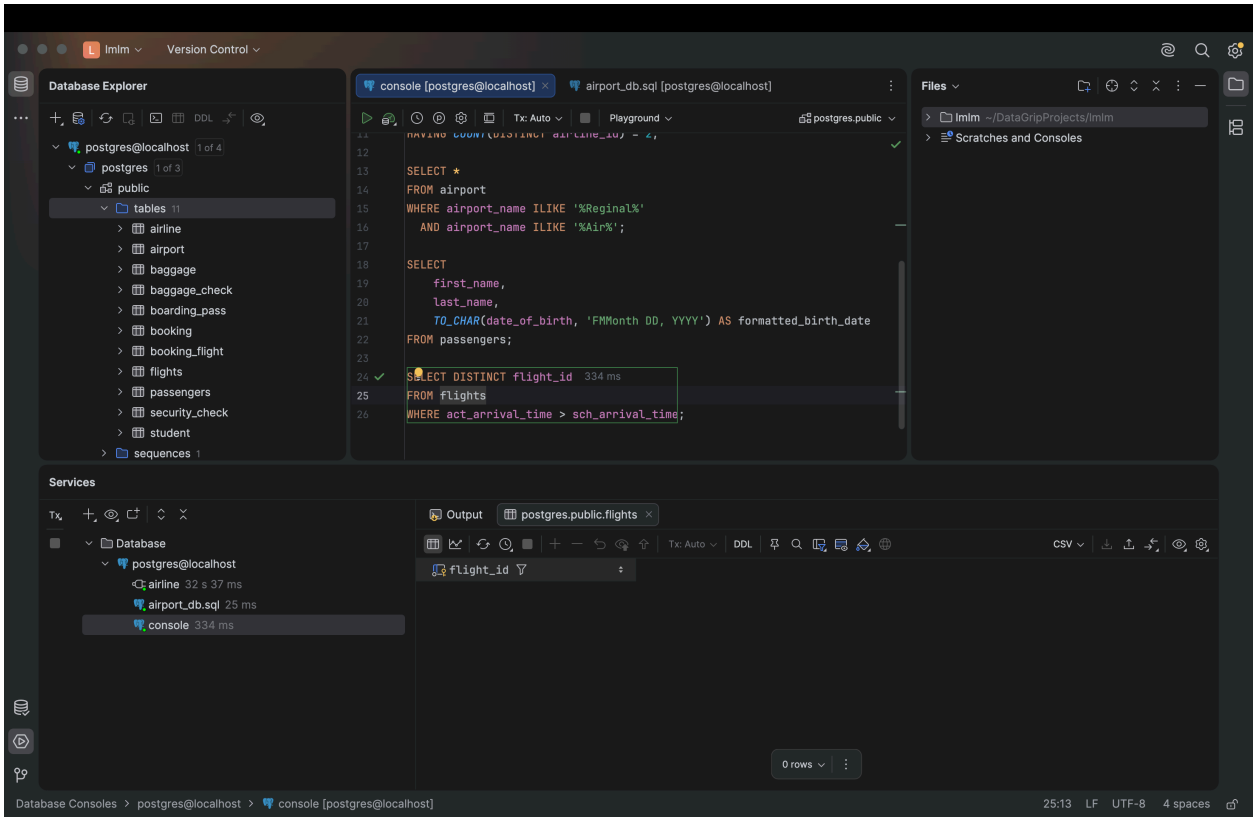


6. Find flight numbers that have been delayed based on the actual arrival time.

SELECT DISTINCT flight_id

FROM flights

WHERE act_arrival_time > sch_arrival_time;

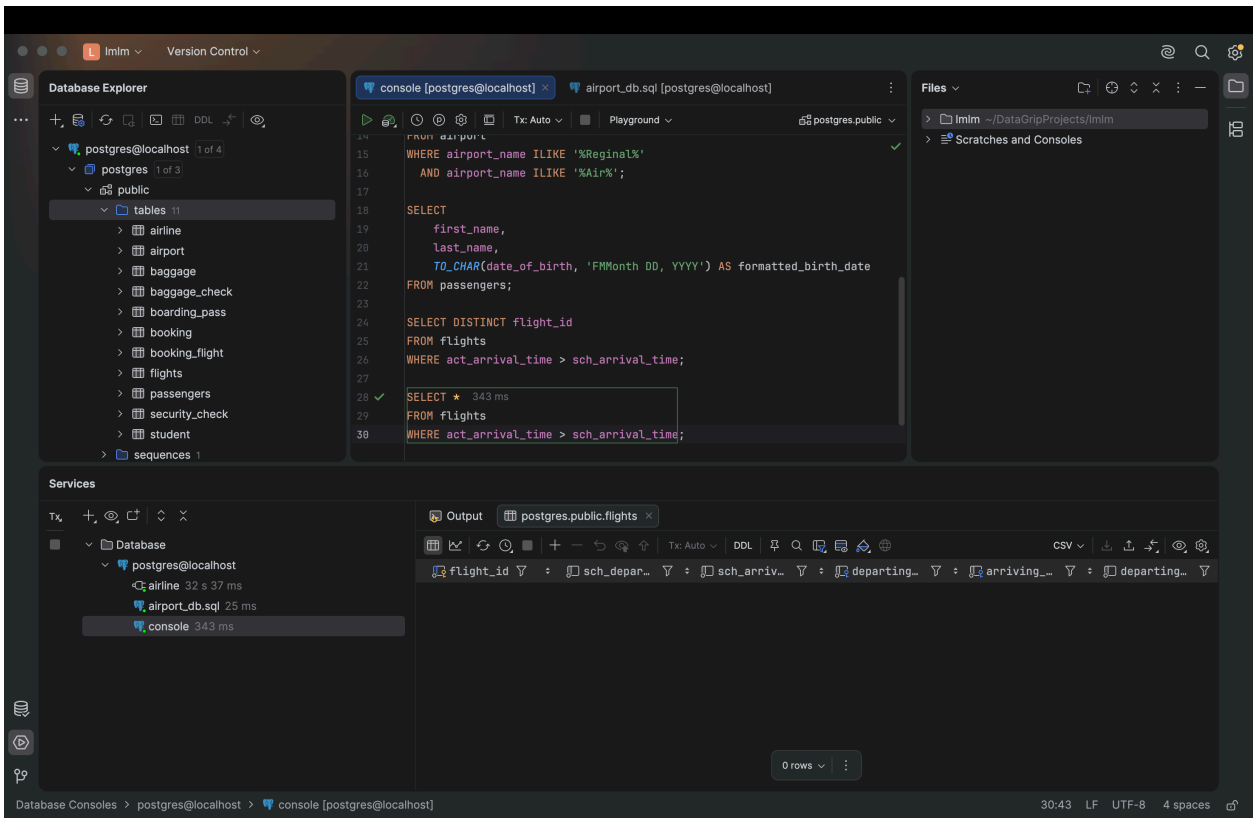


7. Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

SELECT *

FROM flights

WHERE act_arrival_time > sch_arrival_time;



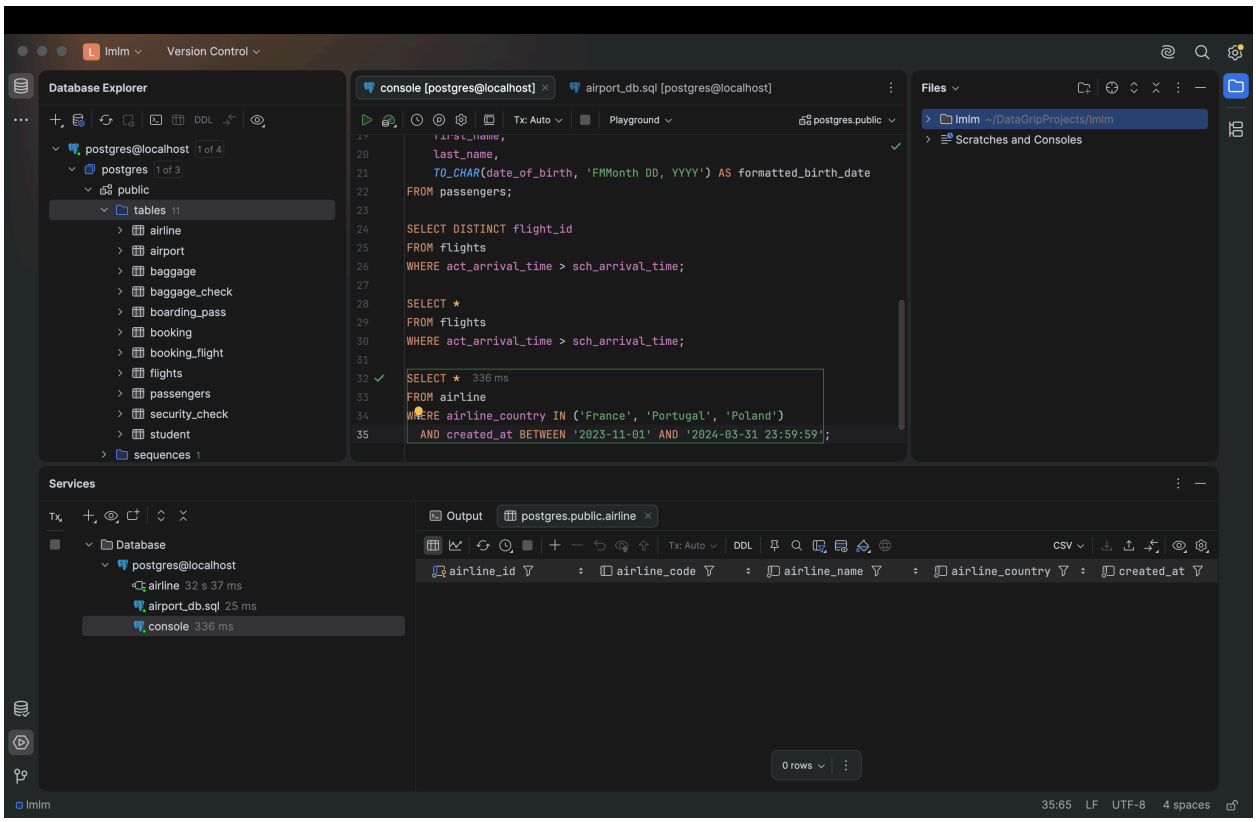
8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

```
SELECT *

FROM airline

WHERE airline_country IN ('France', 'Portugal', 'Poland')

  AND created_at BETWEEN '2023-11-01' AND '2024-03-31 23:59:59';
```



9. Find top3 overweighted baggage with more than 25kg.

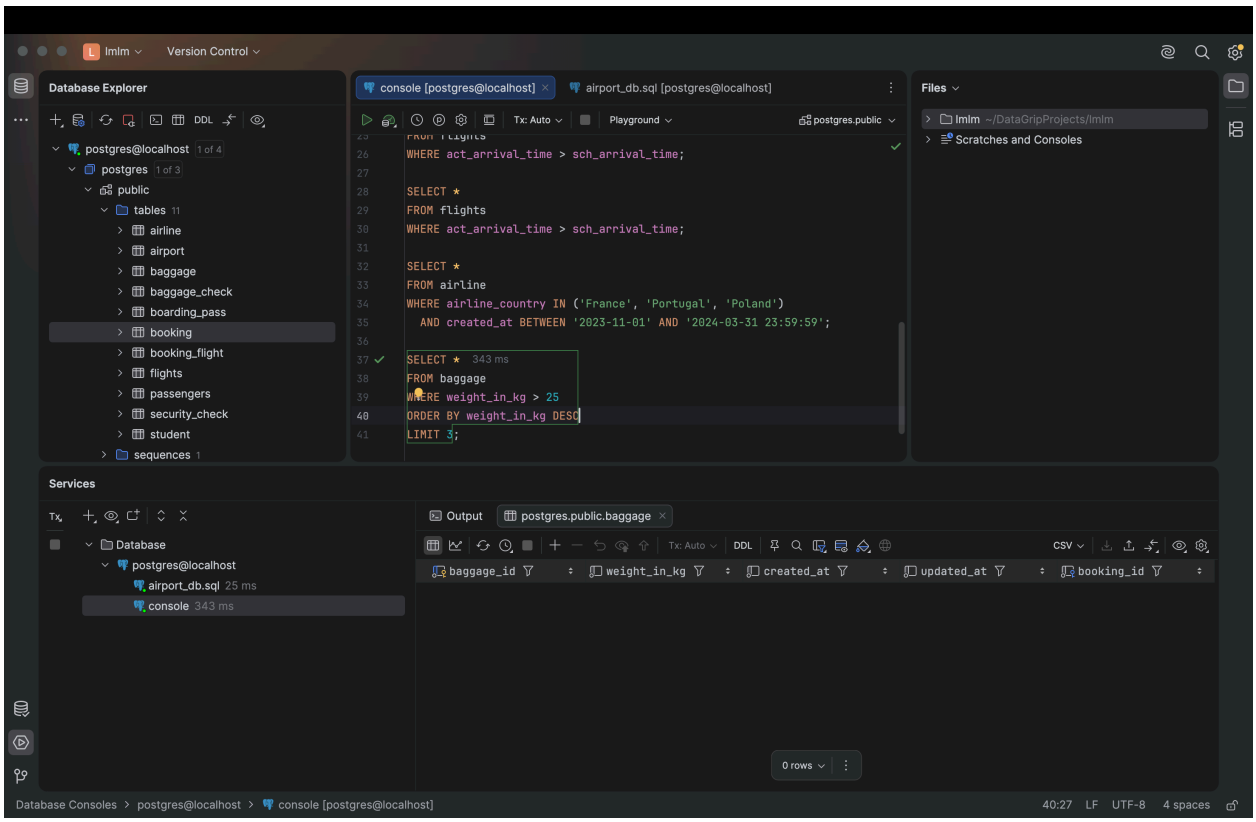
```
SELECT *

FROM baggage

WHERE weight_in_kg > 25

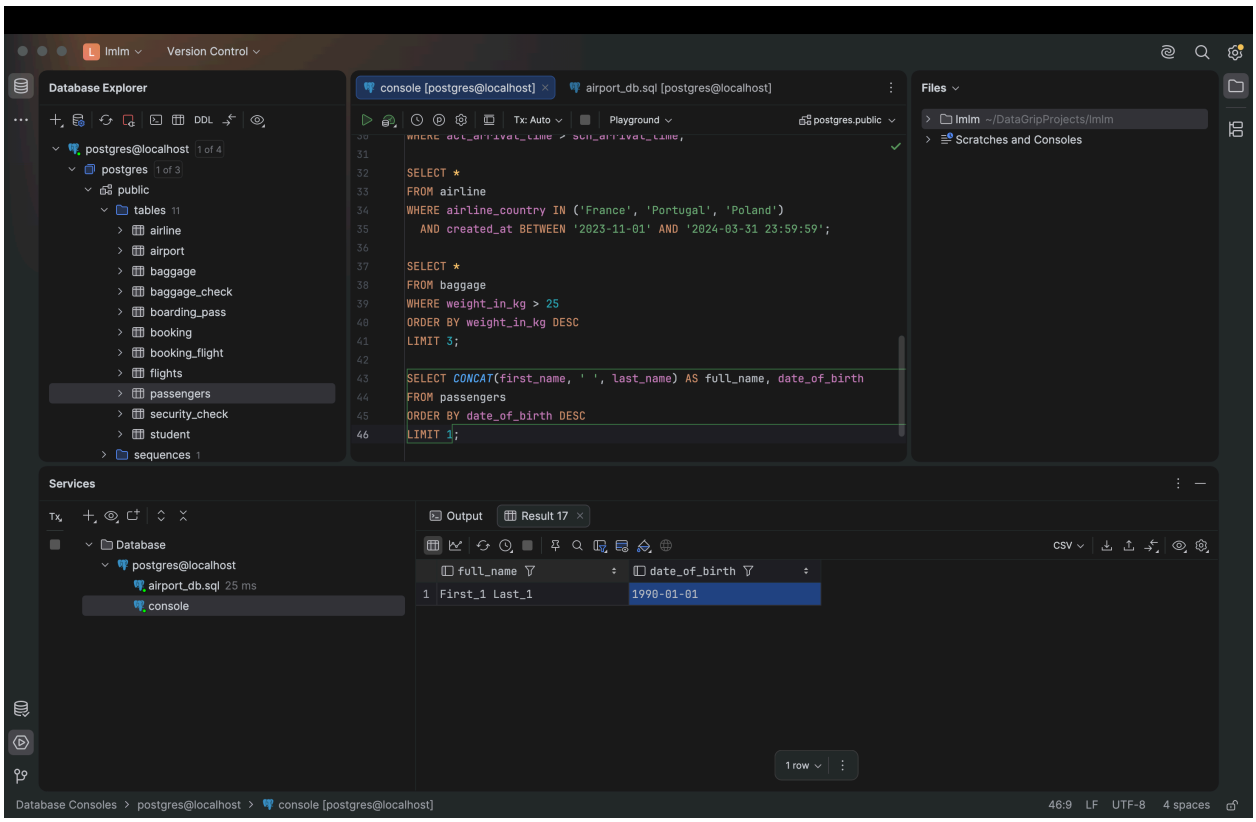
ORDER BY weight_in_kg DESC

LIMIT 3;
```



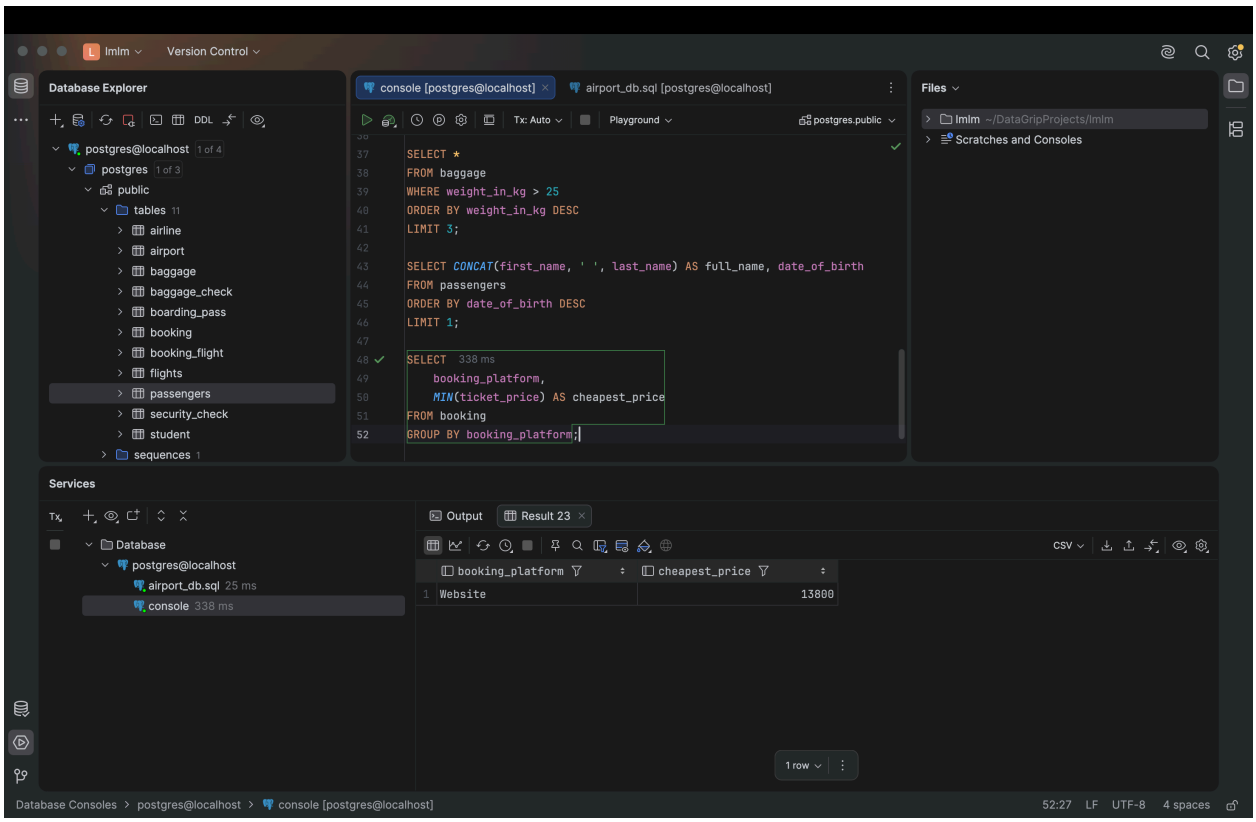
10. Find the youngest passengers' full name.

```
SELECT CONCAT(first_name, ' ', last_name) AS full_name, date_of_birth
FROM passengers
ORDER BY date_of_birth DESC
LIMIT 1;
```



11. Find the cheapest booking price on each booking platform.

```
SELECT
    booking_platform,
    MIN(ticket_price) AS cheapest_price
FROM booking
GROUP BY booking_platform;
```

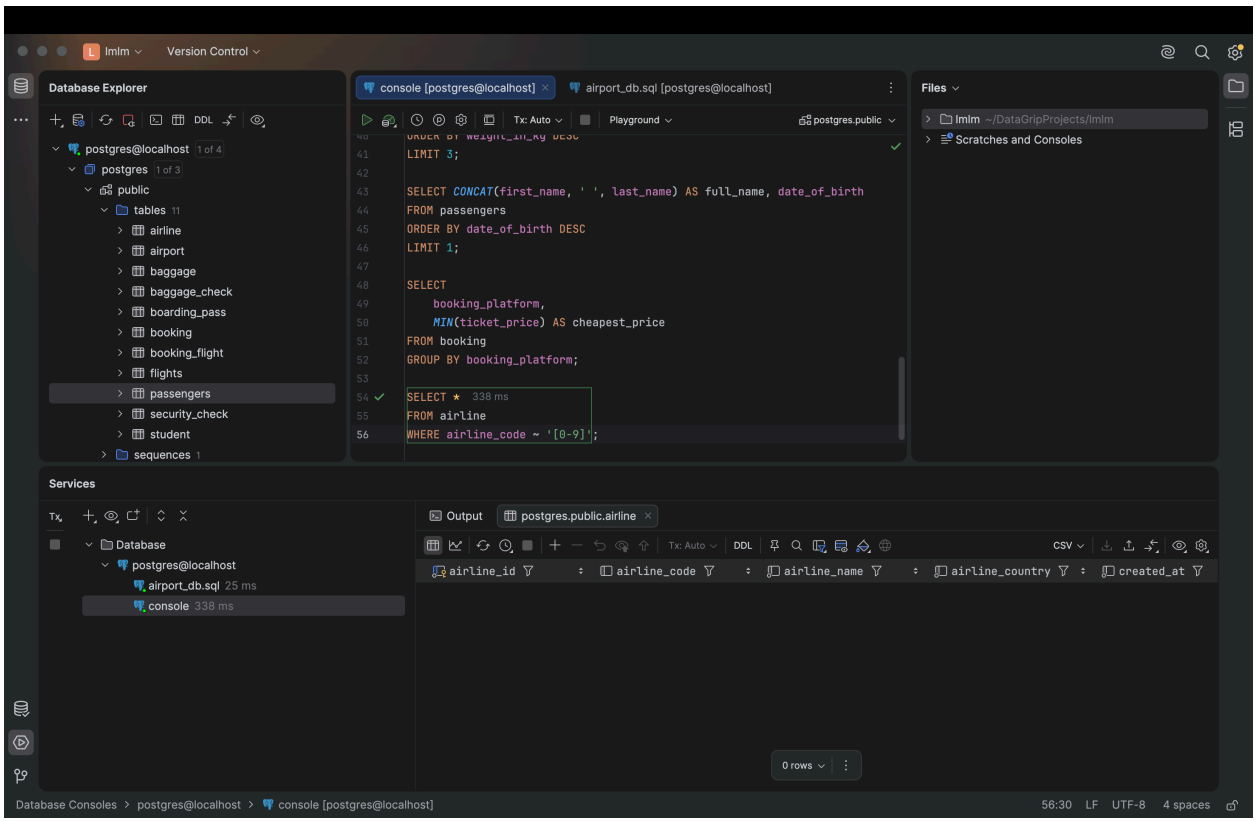


12. Return airlines whose airline_code contains a digit.

```
SELECT *

FROM airline

WHERE airline_code ~ '[0-9]';
```



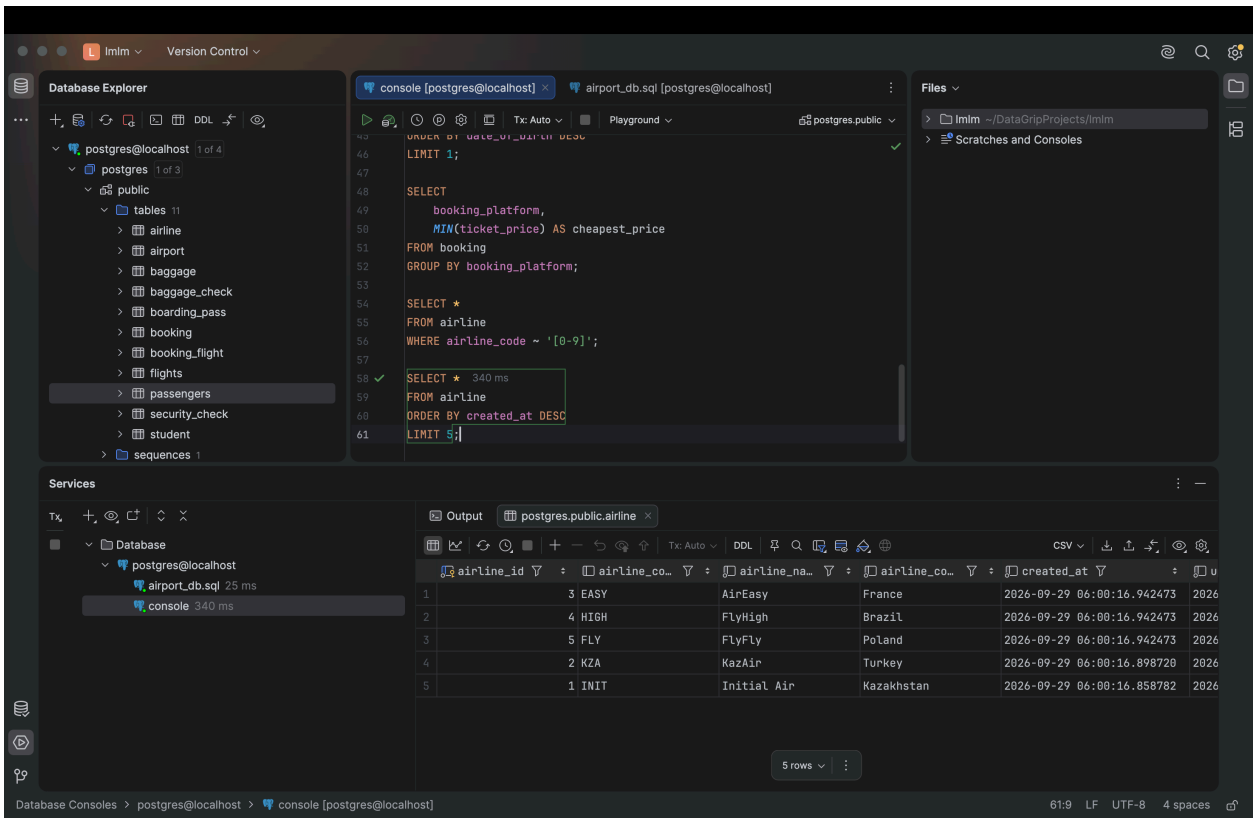
13. List the top5 most recently created airlines.

SELECT *

FROM airline

ORDER BY created_at DESC

LIMIT 5;



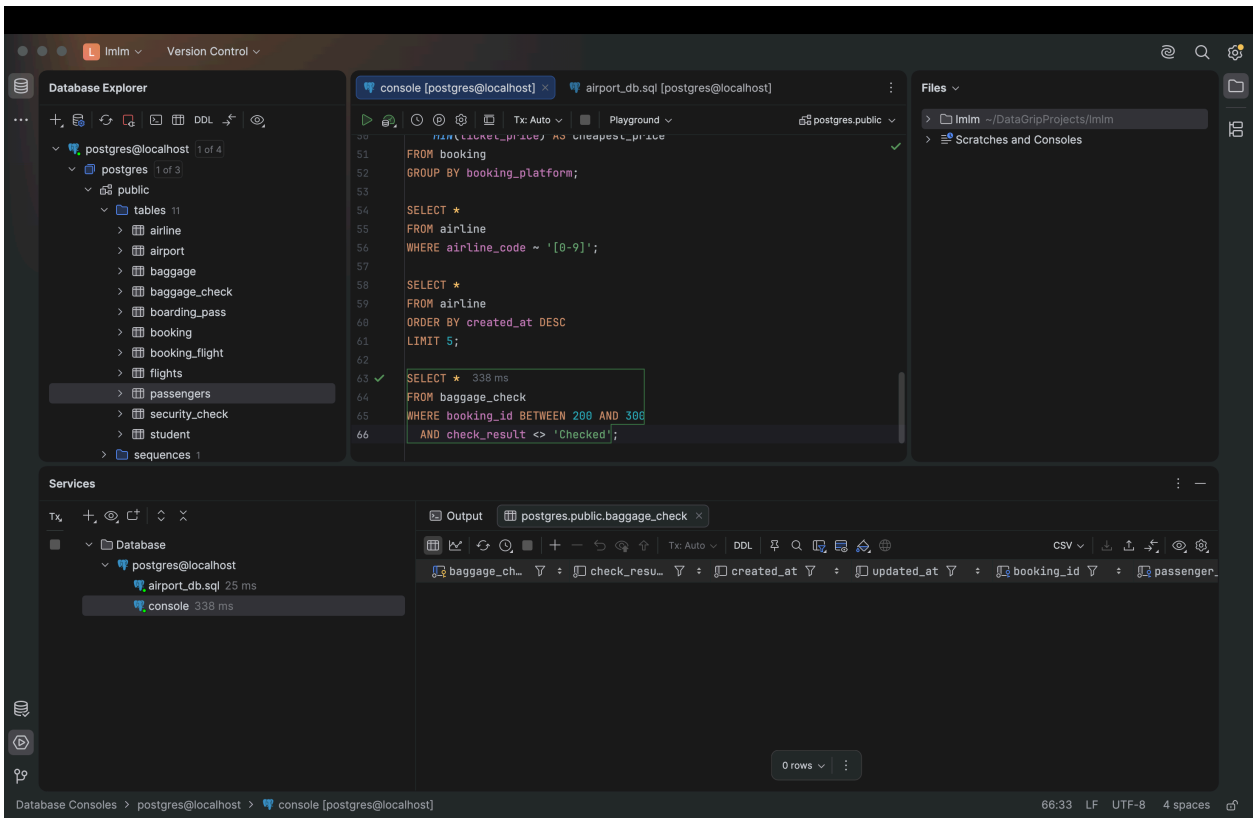
14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

```
SELECT *

FROM baggage_check

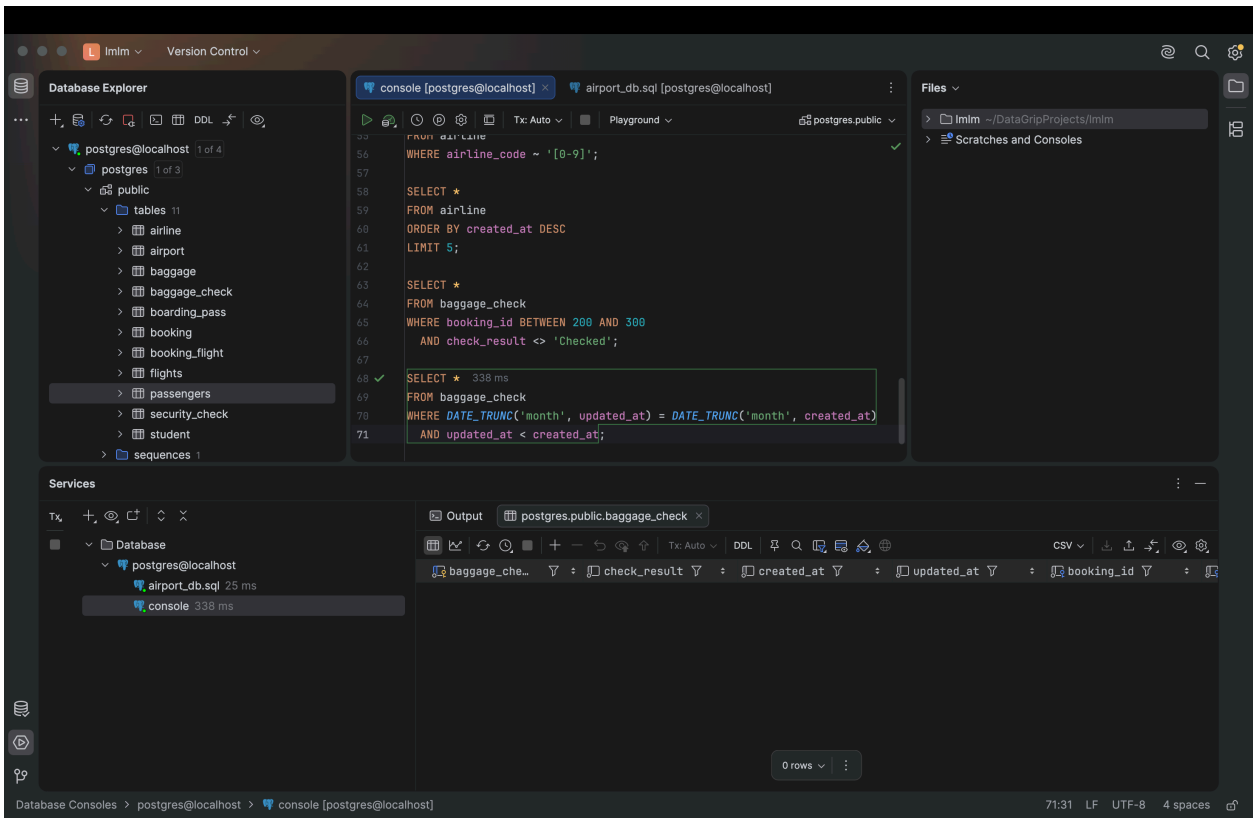
WHERE booking_id BETWEEN 200 AND 300

  AND check_result <> 'Checked';
```



15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

```
SELECT *
FROM baggage_check
WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
  AND updated_at < created_at;
```


lmlm
Version Control
Database Explorer
DDL
postgres@localhost 1 of 4
postgres 1 of 3
public
tables 11
airline
airport
baggage
baggage_check
boarding_pass
booking
booking_flight
flights
passengers
security_check
student
sequences 1
console [postgres@localhost]
airport_db.sql [postgres@localhost]
Tx: Auto
Playground
postgres.public
WHERE airline_code ~ '[0-9]';
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;
SELECT *
FROM baggage_check
WHERE booking_id BETWEEN 200 AND 300
AND check_result <> 'Checked';
SELECT * 338 ms
FROM baggage_check
WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
AND updated_at < created_at;
Files
lmlm ~/DataGripProjects/lmlm
Scratches and Consoles
Services
Database
postgres@localhost
airport_db.sql 25 ms
console 338 ms
Output
postgres.public.baggage_check
DDL
CSV
baggage_che...
check_result
created_at
updated_at
booking_id
0 rows
Database Consoles > postgres@localhost > console [postgres@localhost]
71:31 LF UTF-8 4 spaces